

Sahil Shrikrishna Zele

Data Engineer

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SUMMARY

Data Engineer with 4+ years architecting cloud-native ETL pipelines and real-time streaming solutions across AWS and Azure ecosystems. Specialized in Spark SQL, Kafka event processing, Delta Lake architectures, and Snowflake optimization. Proficient in delivering automated data quality frameworks, accelerating fraud detection systems, and driving substantial cost reductions through performance tuning and cloud migrations.

SKILLS

Methodologies: SDLC, Agile, Waterfall

Programming Languages: Python (Pandas, NumPy, PySpark, scikit learn, Scipy, Matplotlib, Seaborn), SQL, R

Database Technologies: MySQL, Oracle, PostgreSQL, Microsoft SQL Server, MongoDB

Data Pipelines & ETL Tools: Apache Airflow, Talend, Informatica, SSIS, dbt (Data Build Tool)

Cloud Platforms: AWS (S3, EC2, Redshift, Lambda, Kinesis, RDS), Microsoft Azure (Azure Data Lake, Data Factory, Azure Synapse, Azure Functions, Azure Databricks, Azure DevOps)

Big Data Technologies: Apache Spark, Apache Flink, Hadoop, Hive, Kafka

Data Visualization Tools: Tableau, Power BI, Advanced Excel

Version Control & CI/CD: Git, GitHub, Jenkins, Docker, Kubernetes, Terraform, Ansible, Datadog

Soft Skills: Problem-Solving, Effective Communication, Cross-Functional Collaboration, Attention to Detail, Stakeholder Management

EDUCATION

Master of Science in Computer Science

George Mason University, Fairfax, VA

May 2025

Bachelor of Engineering in Computer Engineering

Sinhgad Academy of Engineering, India

May 2019

WORK EXPERIENCE

Data Engineer | PNC, Virginia

January 2025 - Current

- Designed **ETL pipelines** processing 200GB daily financial transactions using **Airflow** and **Spark SQL**, implementing incremental loading strategies and data partitioning that reduced processing time from 5 to 3 hours.
- Built real-time streaming architecture on **Databricks** consuming **Kafka** events at 10K/sec, established **Delta Lake** for ACID-compliant storage with automated schema evolution, enabling faster fraud detection for risk management teams.
- Integrated 3 disparate financial data sources into unified star schema data warehouse using **Spark**, applied SCD Type-2 for historical tracking, supporting executive dashboards with consistent cross-system reporting.
- Implemented automated data quality framework with **Python**-based validation rules and anomaly detection, deployed monitoring dashboards in **Grafana** reducing incident identification time from 2 hours to 15 minutes, preventing downstream errors.
- Optimized **Snowflake** warehouse performance through clustering keys, materialized views, and query rewriting, decreased average query execution time by 60%, achieved 15% cost reduction through workload management and resource rightsizing.
- Led cloud migration of 1TB **SQL Server** databases to **AWS**, architected **S3**-based data lake with **Redshift** analytics layer, automated ingestion using **Lambda** functions ensuring zero data loss during cutover.

Data Engineer | Citus Infotech, India

January 2020 - August 2023

- Executed 3TB enterprise data migration from on-premises **Oracle** to **Azure Synapse** using **Databricks**, designed change data capture patterns for incremental loads, completed migration with 99.9% data accuracy validation.
- Developed comprehensive data quality system in **Azure Data Factory** with schema validation and business rule checks, implemented automated reconciliation processes that reduced data inconsistencies by 30% across production datasets.
- Re-engineered legacy ETL workflows in **Azure Data Factory** using modular pipeline design and parallel processing, improved cloud resource efficiency by 25% while reducing monthly infrastructure costs.
- Built **Spark**-based data pipelines processing 10K daily transactions for risk analytics, created aggregated fact tables and dimensional models supporting ML-powered fraud detection with 25% improved accuracy.
- Strengthened data security posture by implementing column-level encryption and row-level security in **Azure Synapse Analytics**, ensured GDPR compliance for sensitive customer data across 5 business units.
- Automated CI/CD deployment for ETL workflows using **Git** and **Azure DevOps**, integrated **Azure Monitor** with custom alerting logic reducing production incident response time by 50% through proactive issue detection.
- Orchestrated near real-time data ingestion from operational systems to **Azure Data Lake** using **Spark** streaming, transformed batch reporting from 24-hour latency to sub-hour refresh enabling timely business decisions.

PROJECTS

SSL using PySpark:

- Developed a semi-supervised Random Forest model using pseudo-labeling in PySpark, achieving 76% accuracy on the Amazon Reviews Polarity dataset.
- Processed 2M+ reviews with TF-IDF, PCA, and Spark optimizations (partitioning, caching), reducing training time by 40%.
- Evaluated varying labeled/unlabeled splits (e.g., 50/50), improving precision and recall by up to 12% through iterative refinement.

Contextual Vocabulary Enhancement:

- Fine-tuned BERT-Large on SemCor + OMSTI for word sense disambiguation, achieving 66% precision and 0.65 F1-score.
- Implemented cosine similarity-based fallback for rare words, improving generalization accuracy on unseen data by 15%.
- Designed a sentence generation pipeline using BART, GPT-2, and GPT-4 to produce 10,000+ contextual examples.